

MAT 3150 — Homework 1 Solutions

Reading Notes: The Techniques Behind Each Proof

These notes walk through *why* each solution is structured the way it is, not just what it says. The four problems showcase three proof techniques you will use constantly in analysis: **double inclusion** (for set equalities), **induction** (for statements indexed by n), and **contradiction** (for negative statements like “ x is irrational”).

Problem 1: Proving a Set Equality by Double Inclusion

The general strategy

To prove two sets are equal, $S = T$, the standard method (used almost every time you don’t have a slicker algebraic identity available) is *double inclusion*:

$$S = T \iff S \subset T \text{ and } T \subset S.$$

Each inclusion is proved separately, and each inclusion is proved by **element chasing**: to show $S \subset T$, you say “let x be an arbitrary element of S ,” then use only the definition of S to deduce, step by step, that x must also satisfy the definition of T . The word “arbitrary” is doing real work here: because you never used any special property of x beyond “ $x \in S$,” the argument applies to *every* element of S , which is exactly what $S \subset T$ means.

Applying it to $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

Recall $x \in A \setminus B$ means $x \in A$ and $x \notin B$.

Direction 1, $A \setminus (B \cap C) \subset (A \setminus B) \cup (A \setminus C)$. Start from $x \in A \setminus (B \cap C)$, i.e. $x \in A$ and $x \notin B \cap C$. The second fact only tells you x fails to be in *at least one* of B, C — it doesn’t tell you which. This is exactly why the proof splits into cases:

- *Case $x \notin B$:* immediately $x \in A \setminus B$, so certainly x is in the union.
- *Case $x \in B$:* since $x \notin B \cap C$ but $x \in B$, the only way to fail membership in $B \cap C$ is $x \notin C$. So $x \in A \setminus C$, again landing in the union.

Every case lands in $(A \setminus B) \cup (A \setminus C)$, so the inclusion holds. Note the logical skeleton: “not (P and Q)” becomes “not P , or not Q ” (this is De Morgan’s law for logical statements, mirrored by De Morgan’s law for sets: $\complement(B \cap C) = \complement B \cup \complement C$). The case split *is* that law, spelled out by hand instead of quoted as a theorem.

Direction 2, the reverse inclusion. This direction starts from an element of a *union*, which is itself automatically a case split: $x \in (A \setminus B) \cup (A \setminus C)$ means $x \in A \setminus B$ or $x \in A \setminus C$, so the proof again splits into two cases and handles each (using $B \cap C \subset B$, resp. $B \cap C \subset C$, to conclude $x \notin B \cap C$ in either case).

Takeaway. Whenever the defining condition of a set involves “and”/“or”/“not,” expect the proof to case-split along exactly that logical structure. This pattern (not this specific identity) is what you should walk away able to reproduce.

Problems 2 & 3: Proof by Induction

The general strategy

To prove a statement $P(n)$ for all $n \geq n_0$:

1. **Base case.** Verify $P(n_0)$ directly (here $n_0 = 2$ in Problem 2, $n_0 = 1$ in Problem 3).
2. **Inductive step.** Assume $P(n)$ is true (this assumption is called the *inductive hypothesis*) and use it to prove $P(n + 1)$.

If both steps succeed, P holds for every $n \geq n_0$: the base case gives you $P(n_0)$, the inductive step turns that into $P(n_0 + 1)$, then $P(n_0 + 1)$ into $P(n_0 + 2)$, and so on — like a chain of dominoes, each one knocking over the next.

The mechanical trick you’ll use every time: to figure out exactly what $P(n + 1)$ says, take the statement $P(n)$ and substitute $n + 1$ for every occurrence of n . Both solutions do this explicitly (equations (2) and (4) in the two problems) before doing any algebra — do this first, so you know exactly what target you’re proving toward.

Problem 2: a telescoping/product identity

The inductive hypothesis (1) gives you a formula for the product up to the $\frac{1}{n^2}$ term. The product up to the $\frac{1}{(n+1)^2}$ term (the LHS of (2)) is *literally* the LHS of (1) with one extra factor tacked on:

$$\underbrace{\left(1 - \frac{1}{4}\right) \cdots \left(1 - \frac{1}{n^2}\right)}_{\text{this is exactly the LHS of (1)}} \left(1 - \frac{1}{(n+1)^2}\right).$$

This is the key move in almost every induction proof involving a sum or product that grows by one term at a time: *isolate the first n terms so the inductive hypothesis applies directly to them*, then deal with the one new term separately. Once you substitute (1) in for those first n terms, the problem stops being about induction at all and becomes pure algebra: simplify $\frac{n+1}{2n} \left(1 - \frac{1}{(n+1)^2}\right)$ down to $\frac{n+2}{2(n+1)}$, which is exactly the target from (2). The algebra hinges on a difference-of-squares factorization, $1 - \frac{1}{(n+1)^2} = \frac{(n+1)^2 - 1}{(n+1)^2} = \frac{n(n+2)}{(n+1)^2}$ — recognizing $(n+1)^2 - 1 = n(n+2)$ (this is $a^2 - b^2 = (a-b)(a+b)$ with $a = n+1$, $b = 1$) is the one nontrivial algebraic step; everything else is cancellation.

Problem 3: a telescoping sum, via a shift-and-subtract trick

The left side of the $n+1$ statement (equation (4)) is the sum $(2n+3) + (2n+5) + \cdots + (4n+1) + (4n+3)$. Notice this sum *almost* matches the n -statement’s sum $(2n+1) + (2n+3) + \cdots + (4n-1)$, except it’s missing the first term $(2n+1)$ and has two new terms $(4n+1), (4n+3)$ tacked on the end. The solution’s trick is to add and subtract to fix this mismatch:

$$(2n+3) + \cdots + (4n-1) + (4n+1) + (4n+3) = [(2n+1) + (2n+3) + \cdots + (4n-1)] + (4n+1) + (4n+3) - (2n+1).$$

The bracketed term is now *exactly* the LHS of the inductive hypothesis (3), so it can be replaced by $3n^2$. What remains is arithmetic: $3n^2 + (4n+1) + (4n+3) - (2n+1) = 3n^2 + 6n + 3 = 3(n+1)^2$, matching the target. **This “add and subtract the mismatched boundary terms” trick is the general technique for induction on sums where the index range shifts (not just grows by appending one term)** — keep an eye out for it whenever the sum’s starting point also moves with n .

Problem 4: Proof by Contradiction, and Closure Properties

The setup

The key fact used repeatedly is a *closure property*: the rational numbers $\mathbb{Q}$ are closed under addition, subtraction, multiplication, and division (by nonzero rationals) — i.e. combining two rationals with $+$, $-$, $\times$, $\div$ always gives another rational. This is really the whole engine of the proof; everything else is bookkeeping.

The contradiction pattern

To show $x+y$ is irrational (given x rational, y irrational), the solution does *not* attack $x+y$ directly. Instead:

1. **Assume the opposite:** suppose $z = x + y$ is rational.
2. **Derive a consequence:** then $y = z - x$ is a difference of two rationals (z by assumption, x given), hence rational by closure.
3. **Contradict a given fact:** but y was given to be irrational. “ y is rational” and “ y is irrational” can’t both be true — contradiction.
4. **Conclude:** the assumption in step 1 must be false, so $z = x + y$ is irrational.

The same four-step pattern repeats verbatim for xy , using division ($y = w/x$, valid since $x \neq 0$) instead of subtraction. **This is the template for essentially every irrationality proof you’ll see:** assume the rational-looking quantity is rational, algebraically solve for the known-irrational piece, and show that forces it to be rational too.

Why $x \neq 0$ matters

The solution needs to divide by x to isolate $y = w/x$. If x were allowed to be 0, then $xy = 0$ regardless of y , so xy would be rational (namely 0) even though y is irrational — the claim would be false. This is a good example of noticing *why* a hypothesis is stated the way it is, rather than treating it as boilerplate: every hypothesis in a theorem is usually load-bearing.

The final question: is the sum of two irrationals always irrational?

This is a “true or false” question, and to disprove a universal claim (“always”), a single **counterexample** suffices — you don’t need any general argument. $\sqrt{2}$ and $-\sqrt{2}$ are each irrational, but their sum is 0, which is rational. **General principle:** to show a “for all” statement is false, exhibit one instance where it fails; to show an “there exists” statement is true, same idea. This is the opposite of the induction/contradiction techniques above, which prove statements true for *all* n or derive a *general* fact — recognizing which kind of claim you’re faced with (universal vs. existential) tells you which kind of proof to reach for.

Summary: matching technique to claim shape

- **Set equality** $S = T \Rightarrow$ double inclusion, each direction by element-chasing following the logical structure (and/or/not) of the set definitions.
- **“For all $n \geq n_0$ ”** statement $\Rightarrow$ induction: base case, then assume $P(n)$ and prove $P(n+1)$ by isolating the part of $P(n+1)$ that matches $P(n)$ exactly and substituting.
- **“ X is irrational / impossible / not- P ”** statement $\Rightarrow$ contradiction: assume P , derive something that violates a known fact.
- **“Is it always true?” (implicitly, a for-all claim)** $\Rightarrow$ if you suspect it’s false, look for a single counterexample rather than trying to prove a general theorem.